SEAA 4243 FINITE ELEMENT METHOD · 2L + 1T

Plane Stress Part 1 of 3

Two unknowns per node: equilibrium, Hooke’s law and strain–displacement give $[\partial][E][\partial]^T\{u\}^T = -\{f\}$; then the boundary conditions and the element’s [N]

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Text: Mohd Yassin et al., *Two-Dimensional Finite Element Formulations for Heat, Solids and Fluids (with Matlab)*, MSNM 2020 — §3.1 – §3.2.1

ACTIVITY 0 · INDIVIDUAL · RETRIEVAL WARM-UP

Three questions from heat transfer — on paper, no notes

5 MIN

Write down

- The three steps that turn the PDE into $[k]\{\hat{T}\}^T = \{q\} + \{b\}$ (name them).
- What a row of [k] must sum to, and why.
- The matrix form of the heat PDE: $\{\partial\}[E]\{\partial\}^T T = -q_h$. What are the sizes of $\{\partial\}$ and [E]?

WHY Keep question 3 in view: today the same pattern appears with a 2 × 3 operator and a 3 × 3 [E].

5 minutes. Then the outcomes.

LEARNING OUTCOMES · TODAY

By the end of this session you should be able to

1. Derive the plane-stress equilibrium equations from a differential element and write them as $[\partial]\{\sigma\}^T = -\{f\}$.
2. State the plane-stress constitutive law $\{\sigma\}^T = [E]\{\varepsilon\}^T$ and the strain–displacement relation $\{\varepsilon\}^T = [\partial]^T\{u\}^T$, and combine them into $[\partial][E][\partial]^T\{u\}^T = -\{f\}$.
3. Classify traction (natural) and displacement (essential) boundary conditions and write them for a rectangular domain.
4. Write the 2 × 8 shape-function matrix [N] of the four-node element and interpolate u and v at any point.

WHY Text §3.1 – §3.2.1. Next session: [B], [k] and the cantilever worked example; Plane Stress Part 3: stresses, convergence and the project.

SESSION MAP · 180 MINUTES

Six blocks, seven short activities, one break

0:00	0:25	1:00	1:25	1:35	2:05	2:35
Why plane stress?	**Equilibrium → PDE**	**Hooke + strain**	**Break**	**Boundary conditions**	**DOFs and [N]**	**What changes**
25 min	35 min	25 min	10 min	30 min	30 min	25 min
A1 · 5 min	**A2 · 8 min**	**A3 + QR**	no activity	**A4 · 6 min**	**A5 · 8 min**	**Exit ticket**

Revise from solid mechanics

- Stress components σ_{xx}, σ_{yy}, σ_{xy} and the sign convention.
- Hooke's law in 2D; E and ν.
- Engineering shear strain $\varepsilon_{xy} = \partial u/\partial y + \partial v/\partial x$ (the text's symbol for γ_{xy}).

The running example

- Text §3.4: cantilever plate 4 m × 2 m, t = 0.01 m, E = 200 × 10^6 N m^{-2}, ν = 0.33, 5 kN m^{-1} on the free end.
- Today: its boundary conditions and its mesh.
- Next session: its 8 × 8 [k] and solution.

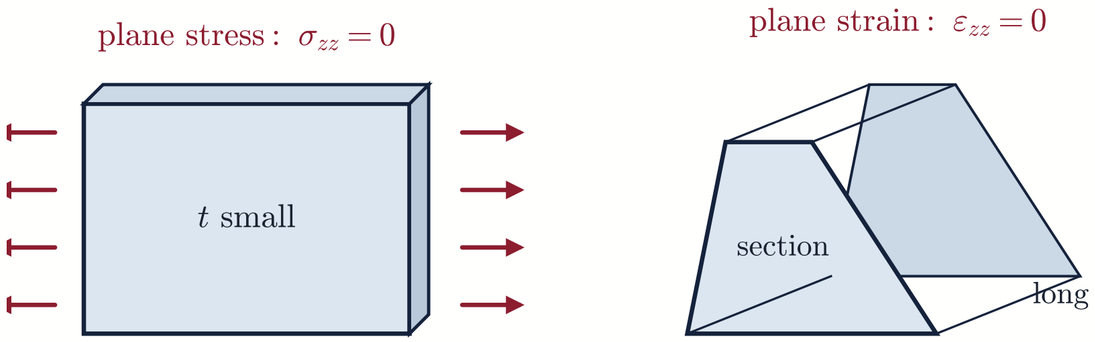
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BLOCK 1 · 0:00 – 0:25

Why plane stress?

Thin bodies loaded in their own plane: shear walls, deep beams, gusset plates, the web of a girder. Two unknowns at every point, u and v.

Thin and loaded in-plane: plane stress. Long with a constant section: plane strain



Plane stress: the thickness t is small and nothing acts on the faces, so $\sigma_{zz} = \sigma_{xz} = \sigma_{yz} = 0$. The body may thin or thicken freely ($\varepsilon_{zz} \neq 0$).

Plane strain: the body is long and every section is loaded the same way, so $\varepsilon_{zz} = 0$ and a stress $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$ develops to prevent it.

Plane stress in practice

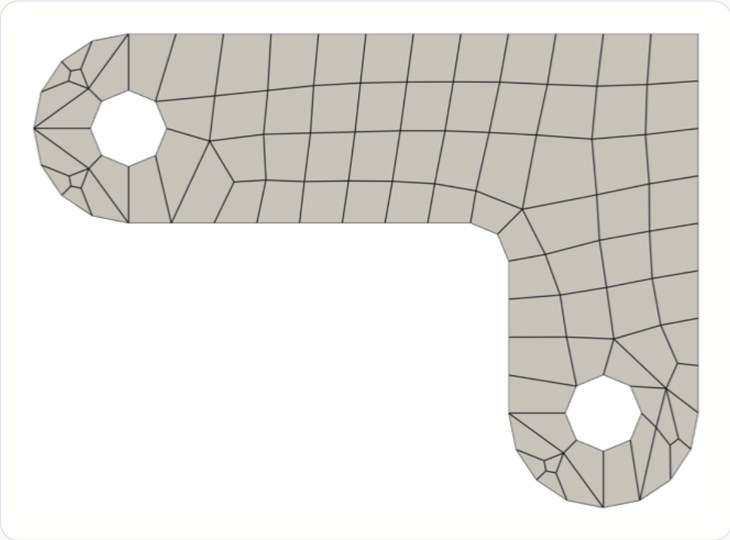
- Shear walls, deep beams
- Gusset plates, connection plates
- Web of a plate girder, thin flanges
- Thin slabs loaded in their plane

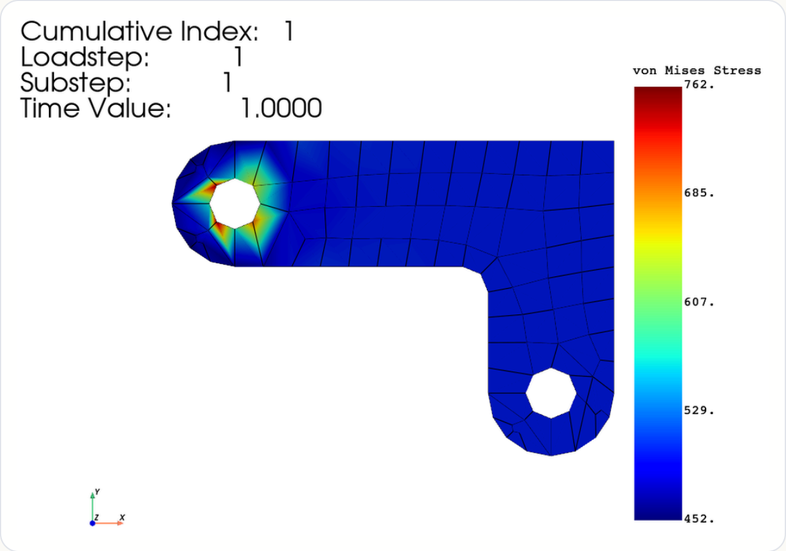
Plane strain in practice

- Gravity dams, long embankments
- Tunnel linings, pipes, culverts
- Retaining walls (per metre run)
- Strip footings

BLOCK 1 · PLANE STRESS IN PRACTICE

A corner bracket in ANSYS, modelled as plane stress

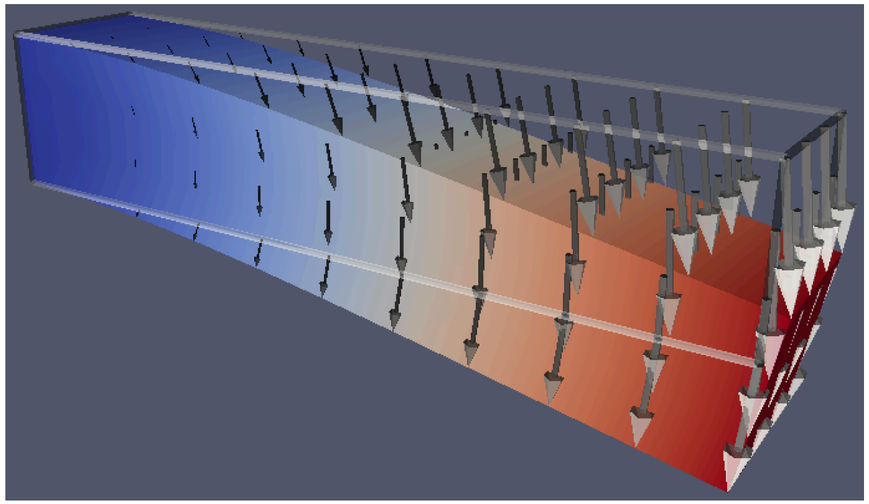




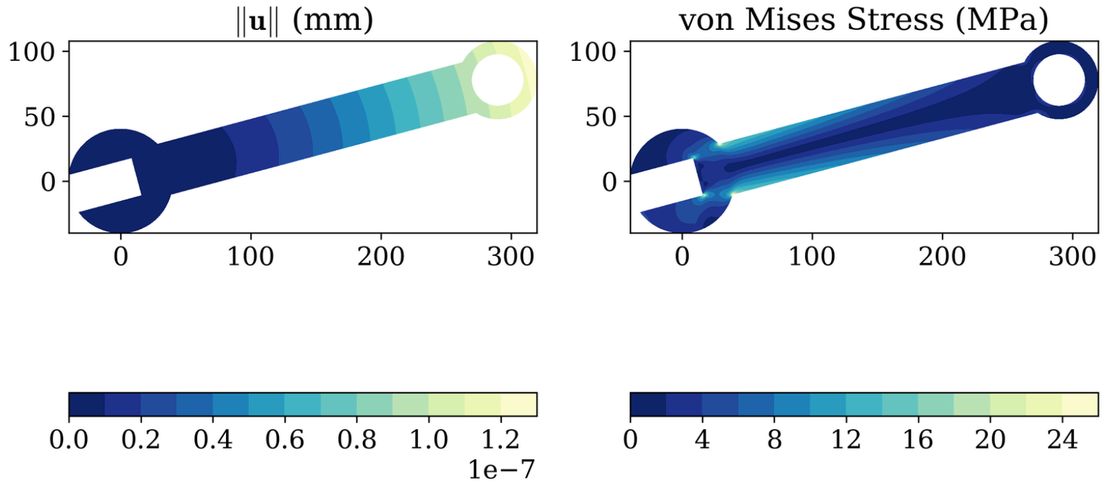
WHY “Because the bracket is thin in the z direction (½-inch thickness) compared to its x and y dimensions, and because the pressure load acts only in the x–y plane, assume plane stress.” — the tutorial’s own justification, word for word the test of this slide.

Images and quotation: PyMAPDL documentation, “Static analysis of a corner bracket” (adapted from the classic Ansys APDL tutorial; PLANE183 elements), © ANSYS, Inc., MIT licence.

The same physics in 3D and in 2D



3D: a clamped beam sagging under its own weight (FEniCS). H. P. Langtangen and A. Logg, *Solving PDEs in Python* (Springer, 2016), CC BY 4.0.



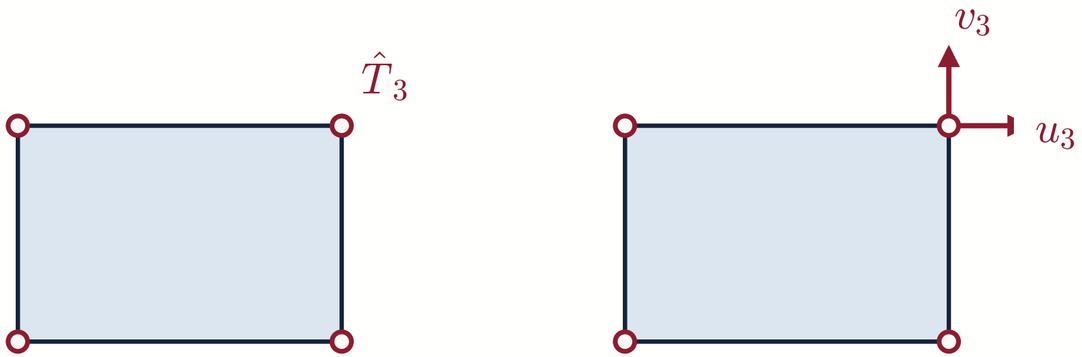
2D: a spanner in plane stress, displacement and von Mises stress. SolidsPy (J. Gómez and N. Guarín-Zapata), MIT licence.

WHY 3D: three displacements u, v, w and six stresses at every point.

CHECK 2D plane stress: u and v, three stresses, $\sigma_{zz} = 0$ because both faces are free. Valid when the body is thin and the loads act in its plane.

BLOCK 1 · THE UNKNOWNS

Displacements u(x, y), v(x, y); from them strains, then stresses



Quantity	Heat transfer	Plane stress
primary field	T (scalar)	u, v (vector)
derived field	flux q_x, q_y	strains ε, stresses σ
material law	Fourier: q = −k∇T	Hooke: {σ} = [E] {ε}
dof per node	1	2
4-node [k]	4 × 4	8 × 8

Heat transfer carried one unknown per node. Plane stress carries two: the nodal displacements u_i and v_i. A 4-node element has 8 degrees of freedom and an 8 × 8 stiffness matrix.

WHY Displacement-based FEM: we solve for u and v, then differentiate to get strains and stresses — so stresses are one order less accurate than displacements.

ACTIVITY 1 · PAIRS · NO CALCULATION

Plane stress, plane strain, or neither? Decide and give the reason

5 MIN

Six structures

- Steel gusset plate at a truss joint, 12 mm thick
- Concrete gravity dam, 300 m long
- Reinforced-concrete shear wall, 200 mm thick, 6 m high
- Circular tunnel lining, 2 km long
- Square slab carrying a lateral floor load (bending)
- Strip footing of a long wall

For each

- Plane stress / plane strain / neither (and why).
- What is zero: σ_{zz} or ε_{zz}?

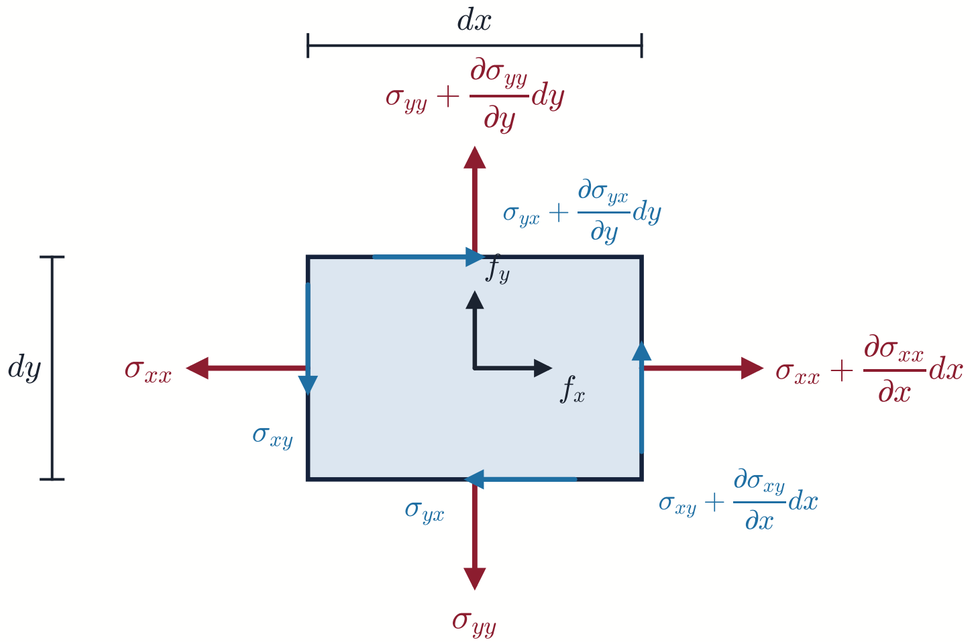
Compare with the pair next to you. We reveal in 5 minutes.

BLOCK 2 · 0:25 – 1:00

2 Equilibrium of a differential element

Sum forces in x and in y; two PDEs in the stresses; write them with the 2 × 3 operator matrix [∂]. Text §3.1.1.

Normal stresses, shear stresses and body forces on a rectangle dx by dy



Dependent variables: σ_{xx}, σ_{yy} (normal) and σ_{xy}, σ_{yx} (shear). Body forces f_x, f_y per unit volume (self-weight: $f_y = -\rho g$).

Far faces carry the near-face stress plus its first-order change — the same Taylor expansion as for the heat flux.

$$\sum F_x = 0, \qquad \sum F_y = 0 \quad (3.1), (3.2)$$

WHY Sign convention: a stress component is positive when it acts in the positive coordinate direction on a face whose outward normal is positive (and vice versa).

BLOCK 2 · FORCE BALANCE IN X · TEXT (3.3)–(3.4)

Forces are stresses times face areas; divide by dx dy

$$\sum F_x = \left(\left(\sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x} dx \right) - \sigma_{xx} \right) dy + \left(\left(\sigma_{yx} + \frac{\partial \sigma_{yx}}{\partial y} dy \right) - \sigma_{yx} \right) dx + f_x \, dx \, dy = 0 \quad (3.3)$$

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + f_x = 0 \quad (3.4)$$

WHY σ_{xx} acts on faces of area dy (times unit thickness); σ_{yx} on faces of area dx. Expand, cancel the equal and opposite near-face terms, divide by dx dy.

$$\left[\frac{\partial \sigma_{xx}}{\partial x} \right] = \frac{\mathrm{N/m^2}}{\mathrm{m}} = \frac{\mathrm{N}}{\mathrm{m^3}} = [\, f_x \,] \qquad \text{(force per unit volume)}$$

ACTIVITY 2 · PAIRS · DERIVATION

Your turn: force balance in y, then the matrix form

8 MIN

Do

- Write $\Sigma F_y = 0$ for the element (which stresses act in y? on which faces?).
- Expand and cancel to get the second PDE.
- Put the two PDEs in matrix form: a 2 × 3 operator matrix times $\{\sigma_{xx}\ \sigma_{yy}\ \sigma_{xy}\}^T$, using $\sigma_{yx} = \sigma_{xy}$.

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + f_x = 0 \quad \text{(3.4) — given}$$

WHY Why $\sigma_{yx} = \sigma_{xy}$? Moment equilibrium about the element centre.

8 minutes. Reveal follows.

ACTIVITY 2 · SOLUTION · TEXT (3.5)–(3.9)

Two equilibrium equations, one operator matrix

$$\sum F_y = \left(\left(\sigma_{yy} + \frac{\partial \sigma_{yy}}{\partial y} dy \right) - \sigma_{yy} \right) dx + \left(\left(\sigma_{xy} + \frac{\partial \sigma_{xy}}{\partial x} dx \right) - \sigma_{xy} \right) dy + f_y \, dx \, dy = 0 \quad (3.5)$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y = 0 \quad (3.6)$$

$$\begin{bmatrix} \dfrac{\partial}{\partial x} & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \begin{Bmatrix} -f_x \\ -f_y \end{Bmatrix} \quad (3.8)$$

$$[\partial]\{\sigma\}^T = -\{f\} \quad (3.9)$$

WHY [∂] is 2 × 3: two equations, three stress components. Its transpose will turn up in the strain–displacement relation.

3

BLOCK 3 · 1:00 – 1:25

Hooke’s law and strain–displacement

Three stresses in terms of three strains; three strains in terms of two displacements. Put together: a PDE in u and v. Text (3.10)–(3.17).

BLOCK 3 · PLANE-STRESS HOOKE'S LAW · TEXT (3.10)–(3.12)

Three stresses from three strains through the 3 × 3 matrix [E]

$$\sigma_{xx} = \frac{E}{1-\nu^2}\,\varepsilon_{xx} + \frac{E\nu}{1-\nu^2}\,\varepsilon_{yy}$$

$$\sigma_{yy} = \frac{E\nu}{1-\nu^2}\,\varepsilon_{xx} + \frac{E}{1-\nu^2}\,\varepsilon_{yy} \qquad (3.10)$$

$$\sigma_{xy} = \frac{E(1-\nu)}{2(1-\nu^2)}\,\varepsilon_{xy}$$

$$\{\sigma\}^T = \begin{Bmatrix}\sigma_{xx}\\ \sigma_{yy}\\ \sigma_{xy}\end{Bmatrix} = \frac{E}{1-\nu^2}\begin{bmatrix}1 & \nu & 0\\ \nu & 1 & 0\\ 0 & 0 & \dfrac{1-\nu}{2}\end{bmatrix}\begin{Bmatrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \varepsilon_{xy}\end{Bmatrix} = [E]\{\varepsilon\}^T \qquad (3.11),\ (3.12)$$

WHY E: Young's modulus; ν: Poisson's ratio. The (1 – ν)/2 entry is G/(E/(1 – ν²)): the shear term uses the engineering shear strain $\varepsilon_{xy} = \gamma_{xy}$.

CHECK Plane strain changes only [E] (next slide). Everything else today is identical.

BLOCK 3 · THE PLANE-STRAIN MATRIX, FOR CONTRAST

Same shape, different constants — and $\varepsilon_{zz} = 0$ instead of $\sigma_{zz} = 0$

$$[E]_{\text{plane strain}} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix}$$

from 3D Hooke's law with ε_zz = 0

Why it matters

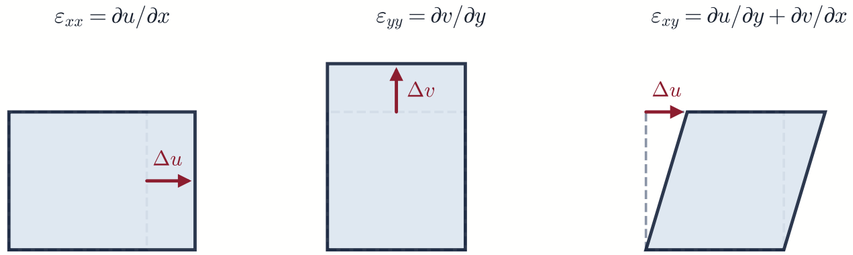
Choosing plane stress for a dam (or plane strain for a thin plate) changes the stiffness by a factor of order $1/(1 - \nu^2)$: several percent for steel, more for rubber-like materials. ANSYS asks you which one; the FEM machinery does not care.

In this course

We formulate and hand-calculate plane stress (text Chapter 3). Plane strain is the same [k] with this [E] substituted — one line in the MATLAB code.

BLOCK 3 · STRAIN–DISPLACEMENT · TEXT (3.13)–(3.15)

Two displacements, three strains, the transpose of [∂]



$$\varepsilon_{xx} = \frac{\partial u}{\partial x}, \qquad \varepsilon_{yy} = \frac{\partial v}{\partial y}, \qquad \varepsilon_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \tag{3.13}$$

$$\{\varepsilon\}^T = \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{Bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \end{Bmatrix} = [\partial]^T \{u\}^T \tag{3.14), (3.15}$$

WHY The 3 × 2 matrix in (3.14) is $[\partial]^T$, the transpose of the equilibrium operator: equilibrium and compatibility are dual.

WATCH OUT ε_{xy} is the engineering shear strain (sum of the two slopes), not the tensor strain (half of it). The text's [E] is built for it.

BLOCK 3 · ASSEMBLE THE PDE · TEXT (3.16)–(3.17)

Substitute strain into Hooke, Hooke into equilibrium: a PDE in u and v

$$\{\sigma\}^T = [E]\,[\partial]^T\{u\}^T \qquad (3.16)$$

$$[\partial]\,[E]\,[\partial]^T\{u\}^T = -\{f\} \qquad (3.17)$$

$$\{\partial\}\,[E]\,\{\partial\}^T T = -q_h \qquad \longleftrightarrow \qquad [\partial]\,[E]\,[\partial]^T\{u\}^T = -\{f\}$$

heat transfer (2.13) ↔ plane stress (3.17)

Read (3.17) from the inside out

- $[\partial]^T\{u\}^T$: displacements → strains (3 × 1).
- $[E][\partial]^T\{u\}^T$: strains → stresses (3 × 1).
- $[\partial][E][\partial]^T\{u\}^T$: stresses → equilibrium (2 × 1), balanced by the body force.

What we will do with it

Exactly what we did with (2.13): substitute $\{u\}^T = [N]\{\hat{u}\}^T$, weight with $[N]^T$, integrate by parts. The boundary term will be the traction (Block 4). Next session.

ACTIVITY 3 · PAIRS · CALCULATION

Evaluate [E] for the cantilever's material

6 MIN

Given (text Fig. 3.4)

- E = 200 × 10^6 N m^{-2}, v = 0.33.
- Plane stress.

Find

- $E/(1 - v^2)$ and the three distinct entries of [E].
- Then: G = E/(2(1 + v)). Does it equal the (3,3) entry?

$$\{\sigma\}^T = \begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{Bmatrix} = [E]\{\varepsilon\}^T$$

CHECK Keep four significant figures; the stiffness matrix the next session is built from these numbers.

6 minutes. Reveal follows.

ACTIVITY 3 · SOLUTION

The constitutive matrix of the cantilever, in $\mathrm{N\,m^{-2}}$

$$\frac{E}{1-\nu^2} = \frac{200\times10^6}{1-0.33^2} = \frac{200\times10^6}{0.8911} = 224.44\times10^6\ \mathrm{N\,m^{-2}}$$

$$= \begin{bmatrix} 224.44 & 74.07 & 0 \\ 74.07 & 224.44 & 0 \\ 0 & 0 & 75.19 \end{bmatrix} \times 10^6\ \mathrm{N\,m^{-2}}$$

$$0.33\times224.44 = 74.07, \qquad \frac{1-0.33}{2}\times224.44 = 75.19$$

CHECK G = E/(2(1 + ν)) = 200 × 10⁶ / 2.66 = 75.19 × 10⁶ N m⁻² — equal to the (3,3) entry, which confirms that ε_{xy} is the engineering shear strain.

QR EXPLORER · PAIRS, ONE PHONE

Plane stress or plane strain? Predict, then check

5 MIN

Scan, then do

1. From the two [E] matrices, predict whether the plate is stiffer in plane strain, and by roughly how much.
2. In the Cantilever Explorer choose 8-node, 8 × 4. Read the tip deflection in plane stress, then in plane strain.
3. Which idealisation suits our 10 mm plate, and which would suit a long dam?

Cantilever Explorer

`razinutm.github.io/My_Lectures/SEAA4243-FEM/labs/cant.html`

5 minutes, optional in class: if the block is running late, set it as homework. Next session you will compute by hand the 2 × 2 result that the Explorer shows.

1:25 – 1:35

Break

Back in 10 minutes. Second half: boundary conditions, degrees of freedom and the shape-function matrix.

BLOCK 4 · 1:35 – 2:05

4 Boundary conditions

Traction (natural) where forces are applied or absent; displacement (essential) where the structure is held. Text §3.1.2.

BLOCK 4 · TWO KINDS OF CONDITION · TEXT (3.18)–(3.23)

On every boundary point, each direction has either a known traction or a known displacement

Neumann · natural · traction specified

$$b_x = \sigma_{xx}\big|_b\, n_x + \sigma_{yx}\big|_b\, n_y, \qquad b_y = \sigma_{xy}\big|_b\, n_x + \sigma_{yy}\big|_b\, n_y \quad \text{(3.18a), (3.19a)}$$

b_x, b_y: force per unit area on the boundary (per unit thickness: force per unit length). A free edge has $b_x = b_y = 0$.

Dirichlet · essential · displacement specified

$$u\big|_b = \bar{u}, \qquad v\big|_b = \bar{v} \quad \text{(3.22), (3.23)}$$

A fixed support: $\bar{u} = \bar{v} = 0$. A roller: one of them.

$$n = (\pm 1,\, 0): \quad b_x = \pm\sigma_{xx}\big|_b, \qquad b_y = \pm\sigma_{xy}\big|_b \quad \text{(3.20)}$$

$$n = (0,\, \pm 1): \quad b_x = \pm\sigma_{yx}\big|_b, \qquad b_y = \pm\sigma_{yy}\big|_b \quad \text{(3.21)}$$

WATCH OUT Each direction separately: on a roller edge u may be fixed (essential) while the traction in y is zero (natural). Count conditions per direction, not per edge.

BLOCK 4 · WHAT FEM DOES WITH EACH KIND

Same table as heat transfer, with forces in place of fluxes

Condition	What is known	Where it enters the FEM	Heat-transfer analogue
Displacement · essential	$\bar{u}$, $\bar{v}$ at nodes	imposed on the assembled system by row–column elimination	$\bar{T}$ imposed
Traction · natural	b_x, b_y on an edge	equivalent nodal loads in {b} from the boundary integral	flux load {b}
Free edge · natural, b = 0	nothing acts	contributes nothing: do nothing	insulated edge
Body force	f_x, f_y per volume	equivalent nodal loads in {q} (e.g. self-weight)	heat generation {q}

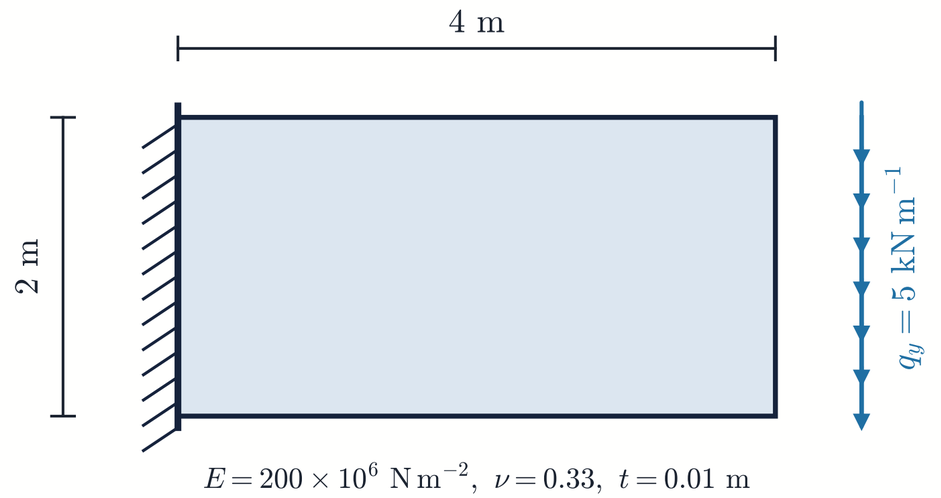
WATCH OUT A free edge is not a support and not a load: it is the do-nothing condition, exactly like an insulated edge.

WHY Reactions at fixed nodes are recovered afterwards from the deleted equations, like boundary fluxes at Dirichlet nodes.

ACTIVITY 4 · PAIRS · CLASSIFICATION

Label the cantilever: each edge, each direction

6 MIN



For the four edges write

- Essential or natural, in x and in y.
- The equation ($\bar{u}$, $\bar{v}$ or b_x, b_y with the outward normal).
- FEM action: impose / load / nothing.

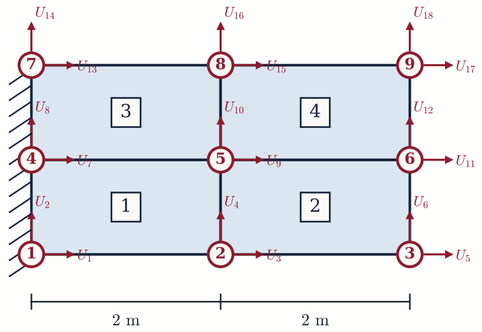
Then: how many of the 18 dofs of the 2 × 2 mesh are known?

6 minutes. Reveal follows.

ACTIVITY 4 · SOLUTION

Six fixed dofs on the left edge, one loaded edge, two free edges

Edge	x-direction	y-direction	FEM action
left (x = 0)	$\bar{u} = 0$, essential	$\bar{v} = 0$, essential	impose $U_1, U_2, U_7, U_8, U_{13}, U_{14} = 0$
right (x = 4)	$b_x = \sigma_{xx} = 0$, natural	$b_y = \sigma_{xy} = -q_y$, natural	nodal loads from the 5 kN m^{-1} traction
top (y = 2)	$b_x = \sigma_{yx} = 0$	$b_y = \sigma_{yy} = 0$	nothing
bottom (y = 0)	$b_x = 0$	$b_y = 0$	nothing



CHECK 18 dofs, 6 known, 12 unknown. Next session: the 12 × 12 reduced system (text (3.48)).

WHY The traction acts downward: $-q_y$ in our sign convention, so the nodal loads come out negative (−2500, −5000, −2500 N in the text).

BLOCK 5 · 2:05 – 2:35

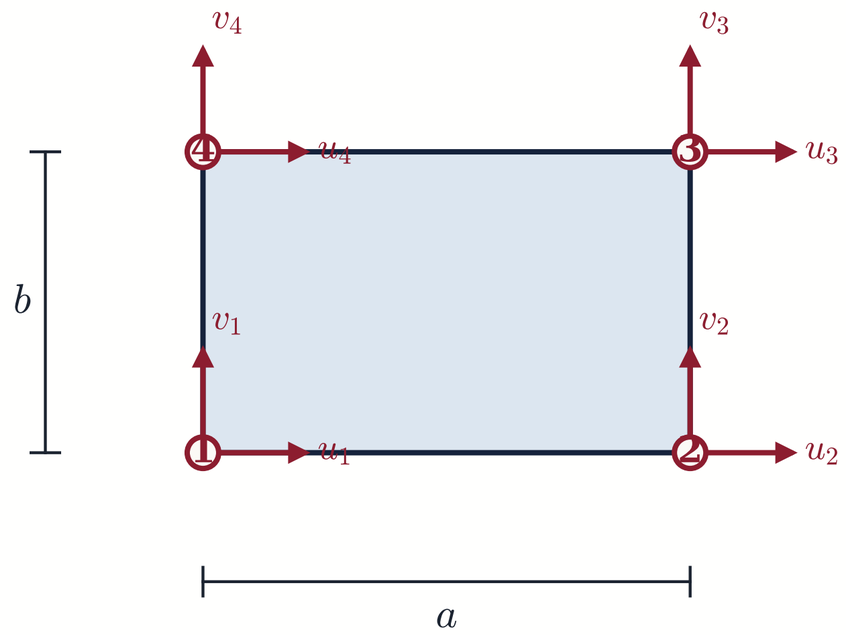
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Degrees of freedom and the shape-function matrix

The same $N_1 \ldots N_4$ as heat transfer, arranged in a 2 × 8 matrix so that both u and v are interpolated at once. Text §3.2.1.

BLOCK 5 · EIGHT DEGREES OF FREEDOM · TEXT FIG. 3.2

Each node carries u (along x) and v (along y); the dof vector interleaves them



$$u = N_1u_1 + N_2u_2 + N_3u_3 + N_4u_4, \qquad v = N_1v_1 + N_2v_2 + N_3v_3 + N_4v_4 \quad (3.24), (3.25)$$

Ordering convention

- $\{\hat{u}\} = \{u_1\ v_1\ u_2\ v_2\ u_3\ v_3\ u_4\ v_4\}$: node by node, u before v.
- Global: node n has dofs U_{2n-1} (x) and U_{2n} (y).
- The same $N_i(x, y)$ as (2.21) interpolate both u and v.

WATCH OUT Keep this ordering everywhere: in [N], in [B], in [k] and in the connectivity. Mixing “all u then all v” with “node by node” is the classic assembly error.

A 2 × 8 matrix: row 1 interpolates u, row 2 interpolates v

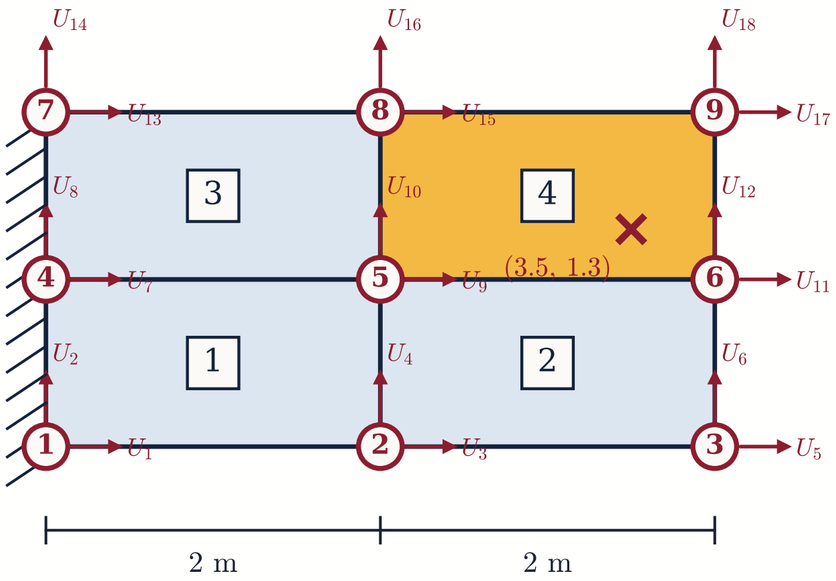
$$\begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{Bmatrix} \quad (3.28)$$

$$\{u\}^T = [N]\{\hat{u}\}^T, \qquad [N]:\ 2 \times 8, \quad \{\hat{u}\}^T:\ 8 \times 1 \quad (3.30)$$

ACTIVITY 5 · PAIRS · CALCULATION

Interpolate u and v at a point of element 4 of the solved cantilever

8 MIN



Given (text (3.49), element 4 = nodes 5, 6, 9, 8)

- (u, v) at node 5: (0, −0.043); node 6: (0, −0.130); node 9: (0.042, −0.131); node 8: (0.031, −0.046) m.
- Element 4: a = 2 m, b = 1 m, its node 1 at global (2, 1).

Find

- Local coordinates of the point (3.5, 1.3) m.
- N_1…N_4 there (you met these numbers in Heat Transfer Part 1).
- u and v at the point.

8 minutes. Reveal follows. Plane Stress Part 3 uses the same point for the stresses.

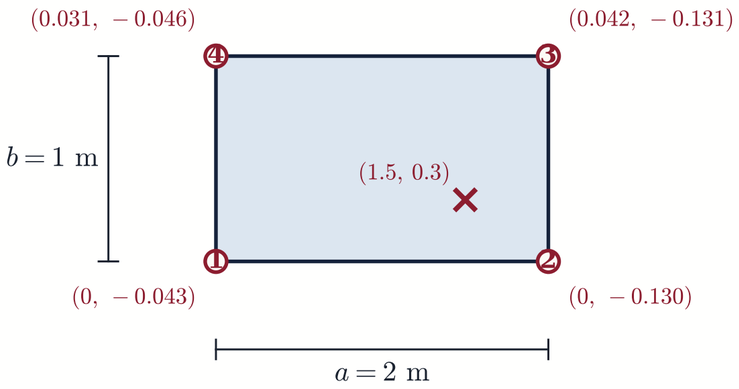
ACTIVITY 5 · SOLUTION

The heat-transfer shape functions, applied to two fields at once

$$a = 2,\ b = 1,\ (x_{\text{loc}}, y_{\text{loc}}) = (1.5,\ 0.3): \qquad \{N\} = \{0.175\ \ 0.525\ \ 0.225\ \ 0.075\}$$

$$u = 0.175(0) + 0.525(0) + 0.225(0.042) + 0.075(0.031) = 0.0118\ \text{m}$$

$$v = 0.175(-0.043) + 0.525(-0.130) + 0.225(-0.131) + 0.075(-0.046) = -0.1087\ \text{m}$$



WHY Node 2 of the element (global node 6, the free end) dominates: half the weight, and the largest v.

CHECK Plane Stress Part 3: differentiate the same interpolation with [B] to get ε and σ at this point.

BLOCK 6 · 2:35 – 3:00

6

The pipeline for plane stress — what changes

Every step repeats; only the sizes grow. Preview of [B] and [k], then the exit ticket.

BLOCK 6 · HEAT TRANSFER AND PLANE STRESS, SIDE BY SIDE

Same seven steps; the objects change size

Step	Heat transfer (Parts 1–2)	Plane stress (Parts 1–3)
PDE	$\{\partial\}[E]\{\partial\}^T T = -q_h$	$[\partial][E][\partial]^T\{u\}^T = -\{f\}$
operator	$\{\partial\}$: 1 × 2	$[\partial]$: 2 × 3
material	$[E] = \mathrm{diag}(k_x, k_y)$: 2 × 2	$[E]$: 3 × 3 (plane stress)
shape functions	$\{N\}$: 1 × 4	$[N]$: 2 × 8
gradient / strain	$\{\partial\}^T\{N\}$: 2 × 4	$[B] = [\partial]^T[N]$: 3 × 8
element matrix	$[k] = \int\{N\}^T\{\partial\}[E]\{\partial\}^T\{N\}\,dA$: 4 × 4	$[k] = t\int[B]^T[E][B]\,dA$: 8 × 8
natural BC	flux $q_b \rightarrow \{b\}$	traction b_x, $b_y \rightarrow \{b\}$
essential BC	$\bar{T}$ at nodes	$\bar{u}$, $\bar{v}$ at nodes

$$\{\varepsilon\}^T = [\partial]^T[N]\{\hat{u}\}^T = [B]\{\hat{u}\}^T, \qquad [B]\ :\ 3 \times 8$$ the next session

$$[k] = t\int_x\int_y [B]^T[E][B]\,dy\,dx \qquad (8 \times 8)$$ the next session

ACTIVITY 6 · INDIVIDUAL · EXIT TICKET

Three questions before you leave

5 MIN

Questions

- A 10 m long culvert loaded uniformly along its length: plane stress or plane strain, and which quantity is zero?
- Write the strain–displacement relation in matrix form and name the matrix.
- For a square element (a = b), what are u and v at the centroid in terms of the nodal values?

$N_i\left(\frac{a}{2}, \frac{b}{2}\right) = \frac{1}{4} \Rightarrow u_{\text{centroid}} = \frac{1}{4}(u_1 + u_2 + u_3 + u_4)$ hint for Q3

$$\{\varepsilon\}^T = \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{Bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \end{Bmatrix} = [\partial]^T \{u\}^T$$

Five minutes. Name and matric number; for feedback, not marks.

WRAP-UP · WHAT TO TAKE AWAY AND WHAT TO PREPARE

Next session: [B], the 8 × 8 [k] and the cantilever solved by hand

Today in four lines

- Equilibrium of the element: $[\partial]\{\sigma\}^T = -\{f\}$ with the 2 × 3 operator $[\partial]$.
- Hooke: $\{\sigma\}^T = [E]\{\varepsilon\}^T$ (3 × 3, plane stress); strain–displacement: $\{\varepsilon\}^T = [\partial]^T\{u\}^T$. Together: $[\partial][E][\partial]^T\{u\}^T = -\{f\}$.
- Traction = natural (load or nothing); displacement = essential (imposed). Count per direction.
- $\{u\}^T = [N]\{\hat{u}\}^T$ with [N] 2 × 8, dofs ordered node by node, u before v.

Before the next session

- Assignment 4 due the next session.
- Read text §3.2.2 – §3.3 (pages 89–96): the k_{34} steps and $s_1 \ldots s_{12}$.
- Redo Activity 3 for plane strain with the matrix on the contrast slide.
- Bring the heat-transfer MATLAB script: the next session we modify it for plane stress.

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